

Gum Disease and the Cell Danger Response in Alzheimer's Disease

Gingivalis - The same bacteria responsible for gum disease (gingivitis) are toxic to the brain. An abundance of this bacteria can generate toxic enzymes (gingipains), which can reach the brain and trigger dementia symptoms.

In a healthy brain

- **Toxins arrive:** Gingipain toxins reach the brain.
- **Immune Response:** The brain begins a cell danger response and deploys amyloid peptides as a sticky “net” to trap and neutralize the toxins.
- **Cleanup:** Brain microglia (the immune system’s janitor cells) digest the amyloid-trapped debris.
- **Restoration:** Debris are flushed out via brain fluid during deep sleep.
- **Baseline:** The brain returns to normal operations.

In a low energy brain

- **Toxins arrive:** Gingipain toxins reach the brain.
- **Immune Response:** The brain begins a cell danger response and deploys amyloid proteins as a sticky “net” to trap and neutralize the toxins.
- **Failure to Clean:** Because the brain lacks energy, brain microglia (janitor cells) are too sluggish to digest the debris.
- **Accumulation:** The trapped toxins and amyloid remain stagnant and build up in the brain, clumping together into plaques.
- **Chronic Loop:** The cell danger response remains stuck in a chronic, damaging loop.

The bottom line

Gum disease can be a trigger for Alzheimer's disease in a low-energy brain. Maintaining an oral hygiene routine focused on raising the pH of the mouth (more alkaline) and breaking up plaque (biofilm) significantly reduces the toxic burden on the brain, which can prevent the worsening of cognitive decline.

By keeping your mouth healthy, you can prevent the worsening of cognitive decline and give the brain a break from toxins, freeing up its resources for rest, repair, and daily activity.

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